CSIT 491 — Lab 2

VLAN Configuration and Network Segmentation

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Course: CSIT 491

Instructor: Ramy Othman

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1. Introduction

The purpose of this lab was to configure Virtual Local Area Networks (VLANs) and implement network segmentation using Cisco Packet Tracer. Two VLANs were created to separate network traffic between Students (VLAN 10) and Faculty (VLAN 20).

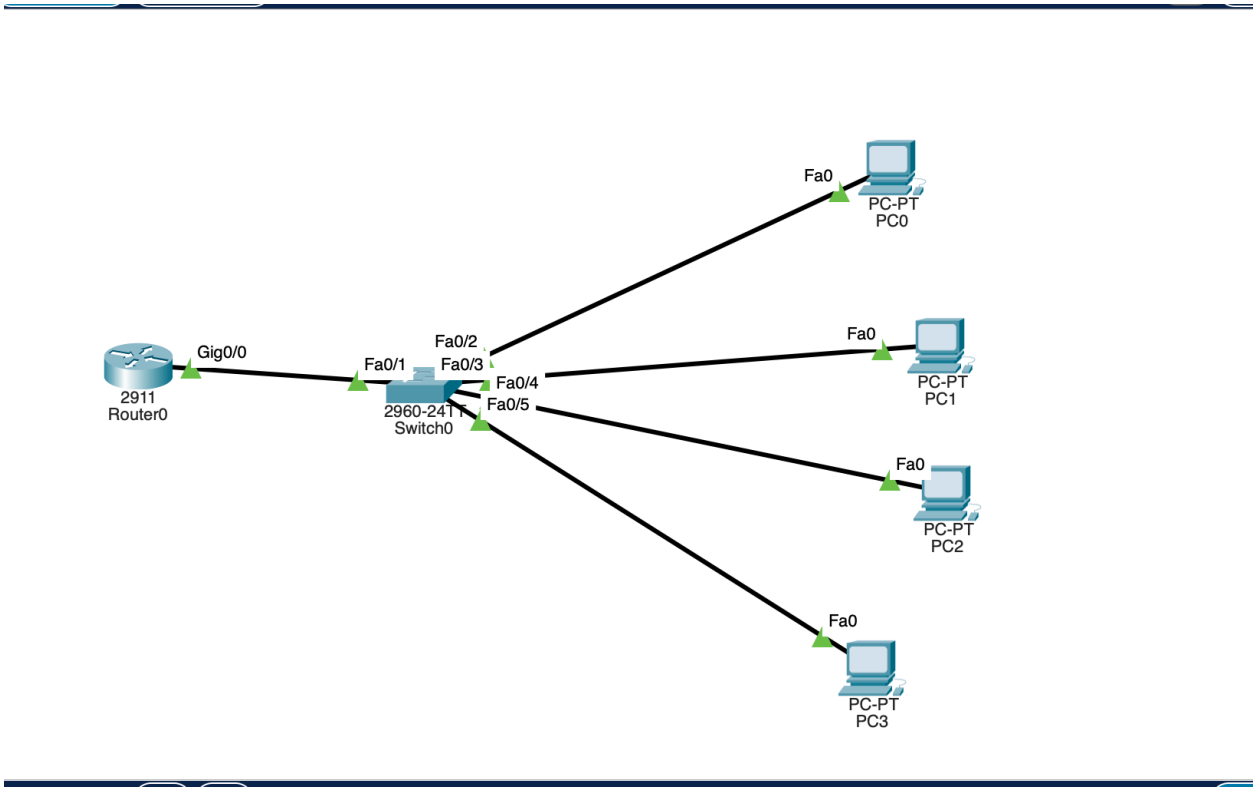
Inter-VLAN routing was enabled using a router-on-a-stick configuration with subinterfaces and IEEE 802.1Q encapsulation. Connectivity was verified through successful and failed ping tests.

This lab demonstrates understanding of:

- VLAN creation and port assignment
- Trunk configuration
- Router subinterfaces
- Inter-VLAN routing
- Connectivity verification

2. Network Topology

The network consisted of 1 Router (Cisco 2911), 1 Switch (Cisco 2960-24TT), and 4 PCs. The router was connected to the switch using a trunk link on Fa0/1.



VLAN Assignments

VLAN	Name	Ports	IP Network
10	Students	Fa0/2 – Fa0/3	192.168.10.0/24
20	Faculty	Fa0/4 – Fa0/5	192.168.20.0/24

3. VLAN Configuration on Switch

The following commands were used to create the VLANs and assign ports:

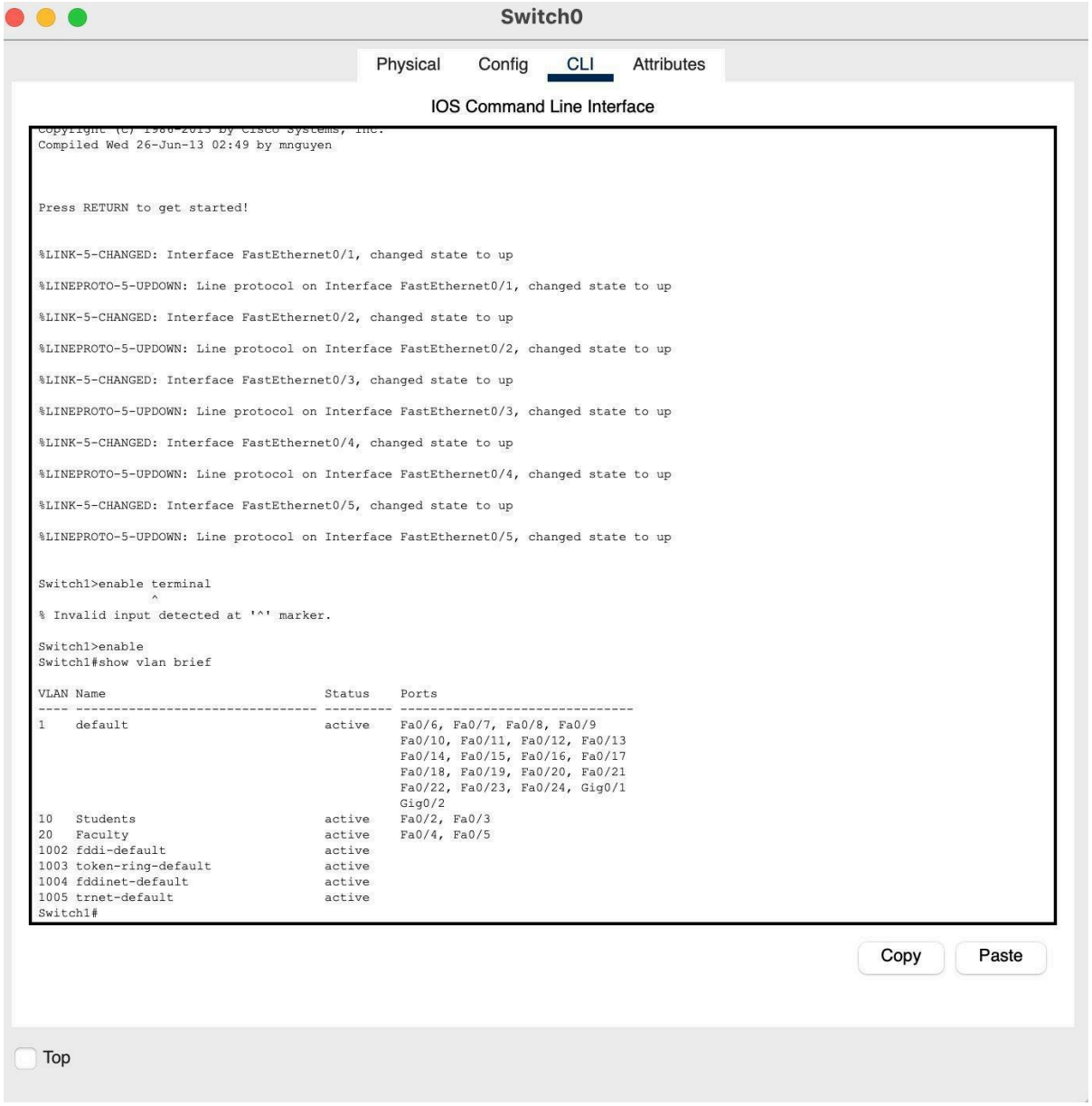
```
enable
configure terminal

vlan 10
 name Students
exit

vlan 20
 name Faculty
exit

! Students port assignment
interface range fa0/2-3
 switchport mode access
 switchport access vlan 10
exit

! Faculty port assignment
interface range fa0/4-5
 switchport mode access
 switchport access vlan 20
exit
```



4. Trunk Configuration

To allow multiple VLANs to traverse the link between the switch and router, the switch port connected to the router was set to trunk mode:

```
interface fa0/1
 switchport mode trunk
exit

show interface trunk
```

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up


Switch1>enable terminal
                ^
% Invalid input detected at '^' marker.

Switch1>enable
Switch1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Students                         active    Fa0/2, Fa0/3
20   Faculty                          active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch1#
Switch1#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20

Switch1#
```

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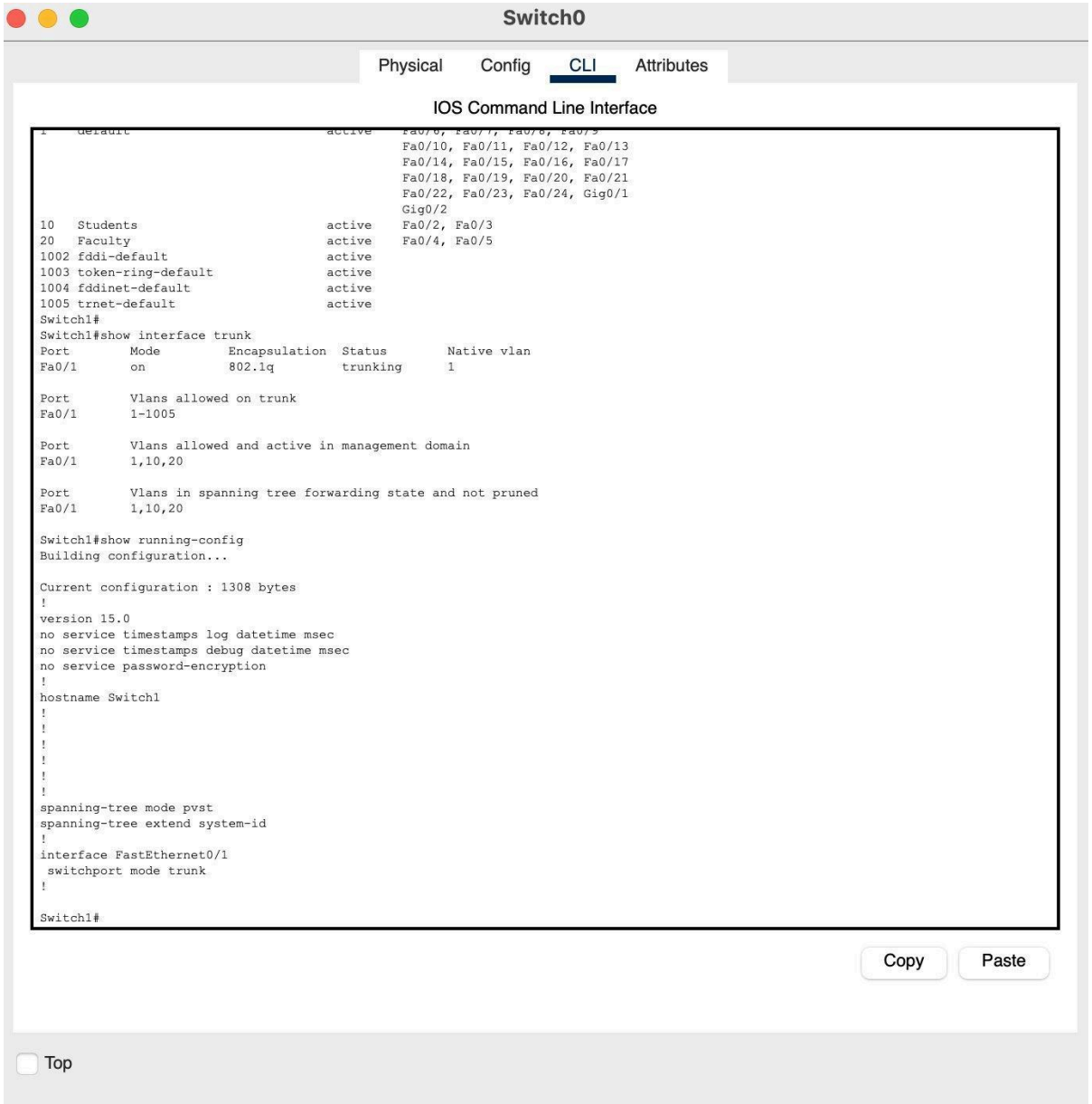
Top



5. Router Configuration

Subinterfaces were configured on the router to enable inter-VLAN routing. Each subinterface was assigned to a VLAN using 802.1Q encapsulation:

```
enable
configure terminal

interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
```

```
interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit

show ip route
show ip interface brief
```

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up


R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20

R1#
```

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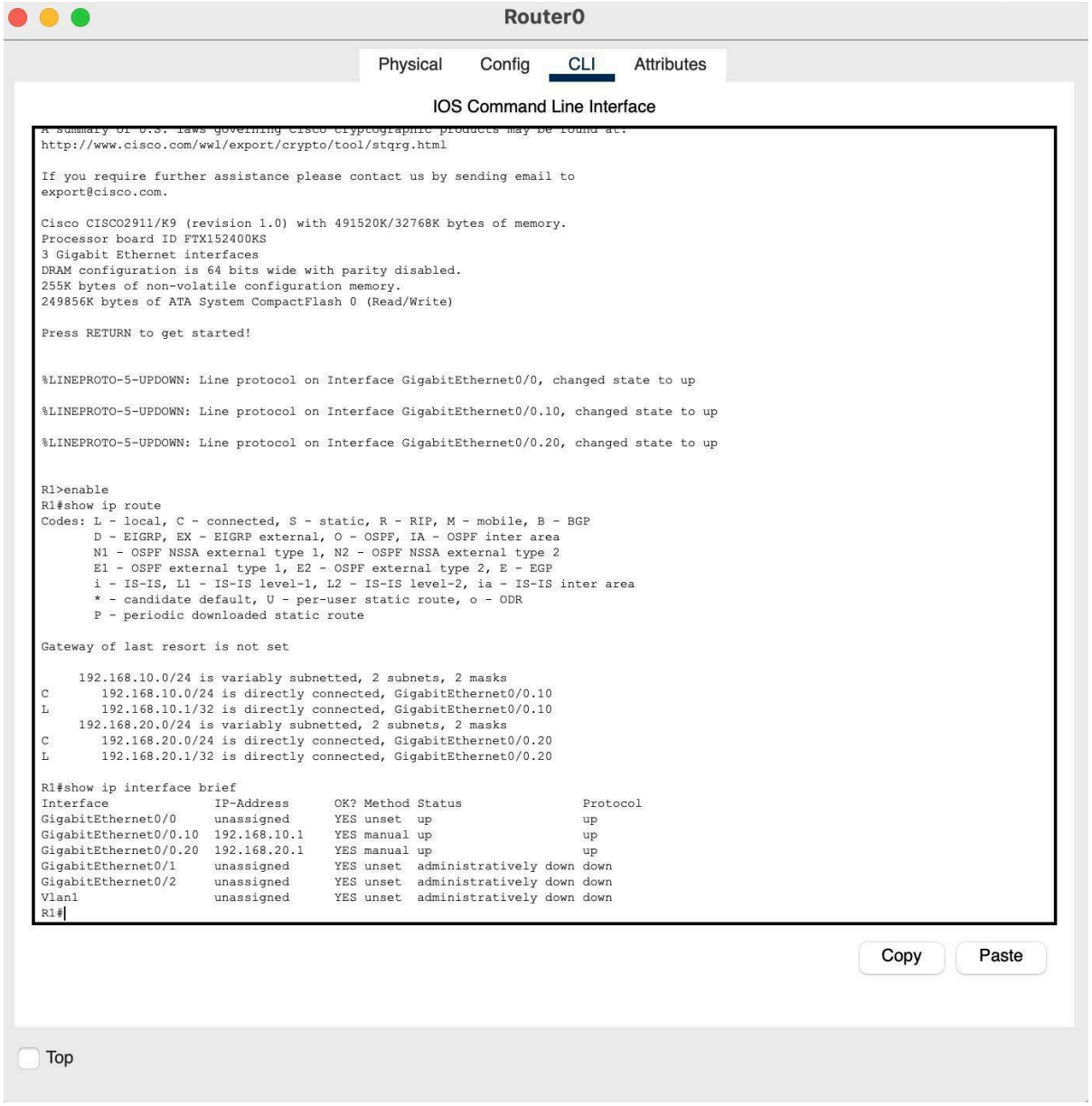
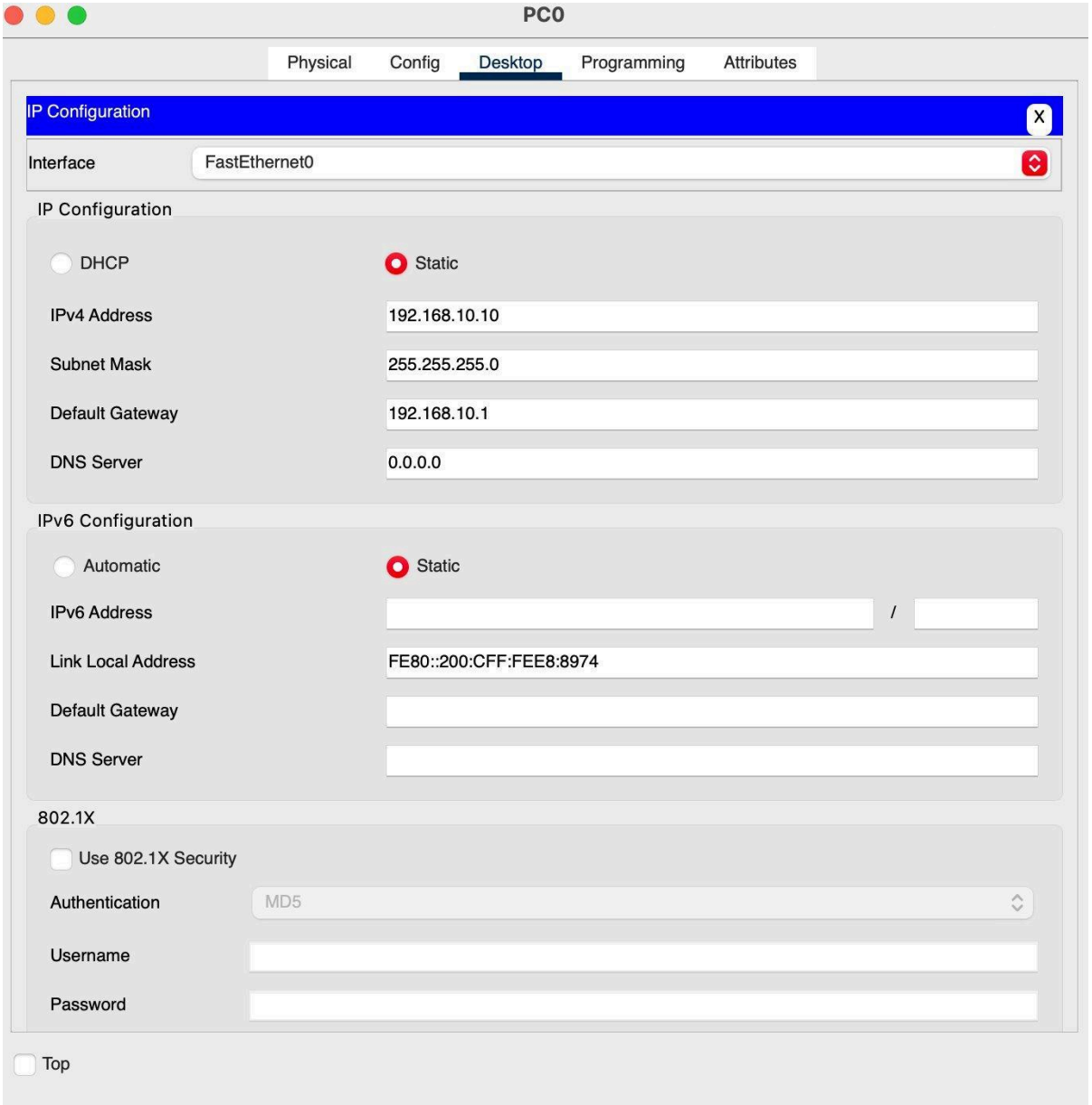
Top

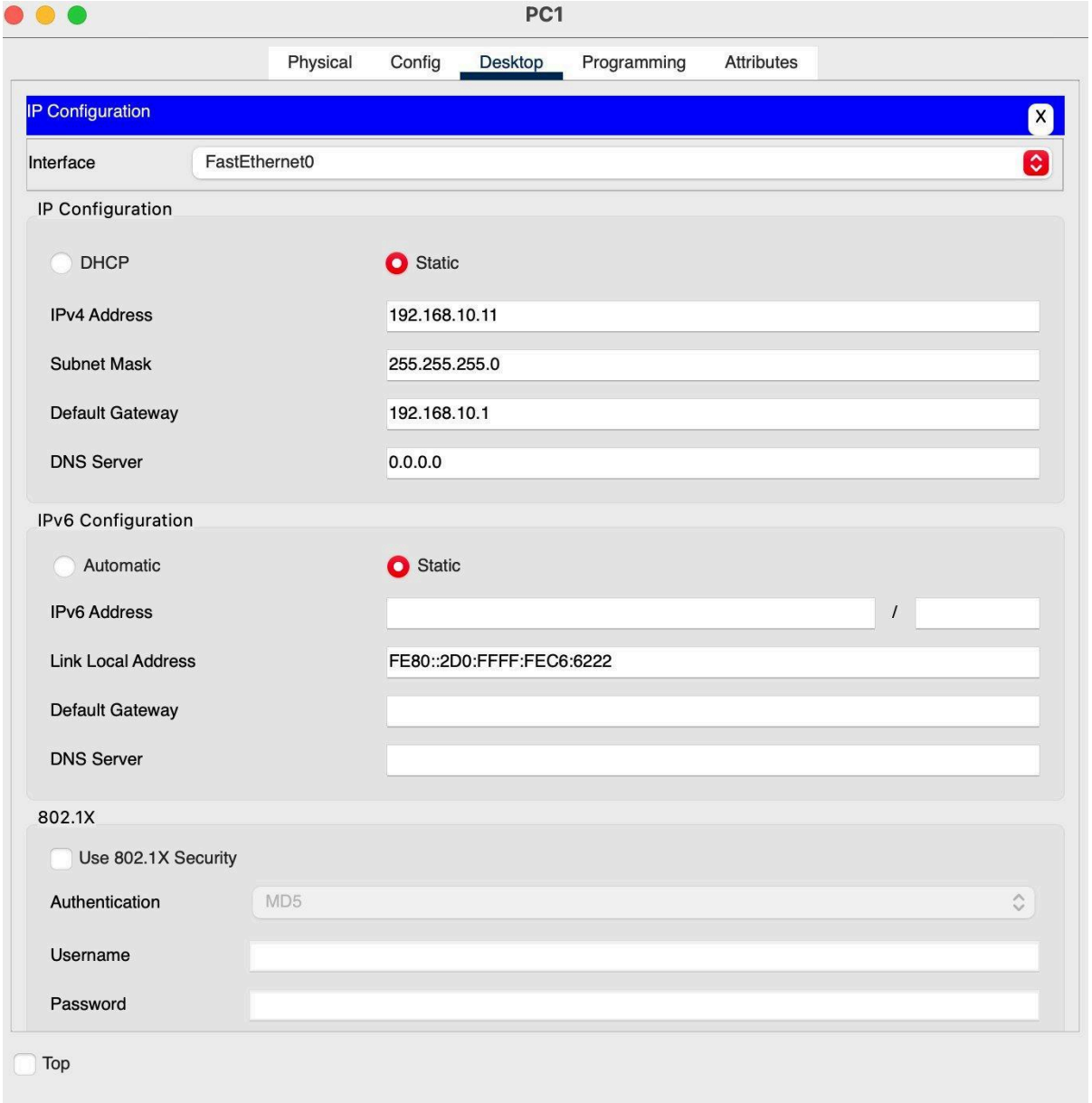


6. PC IP Configuration

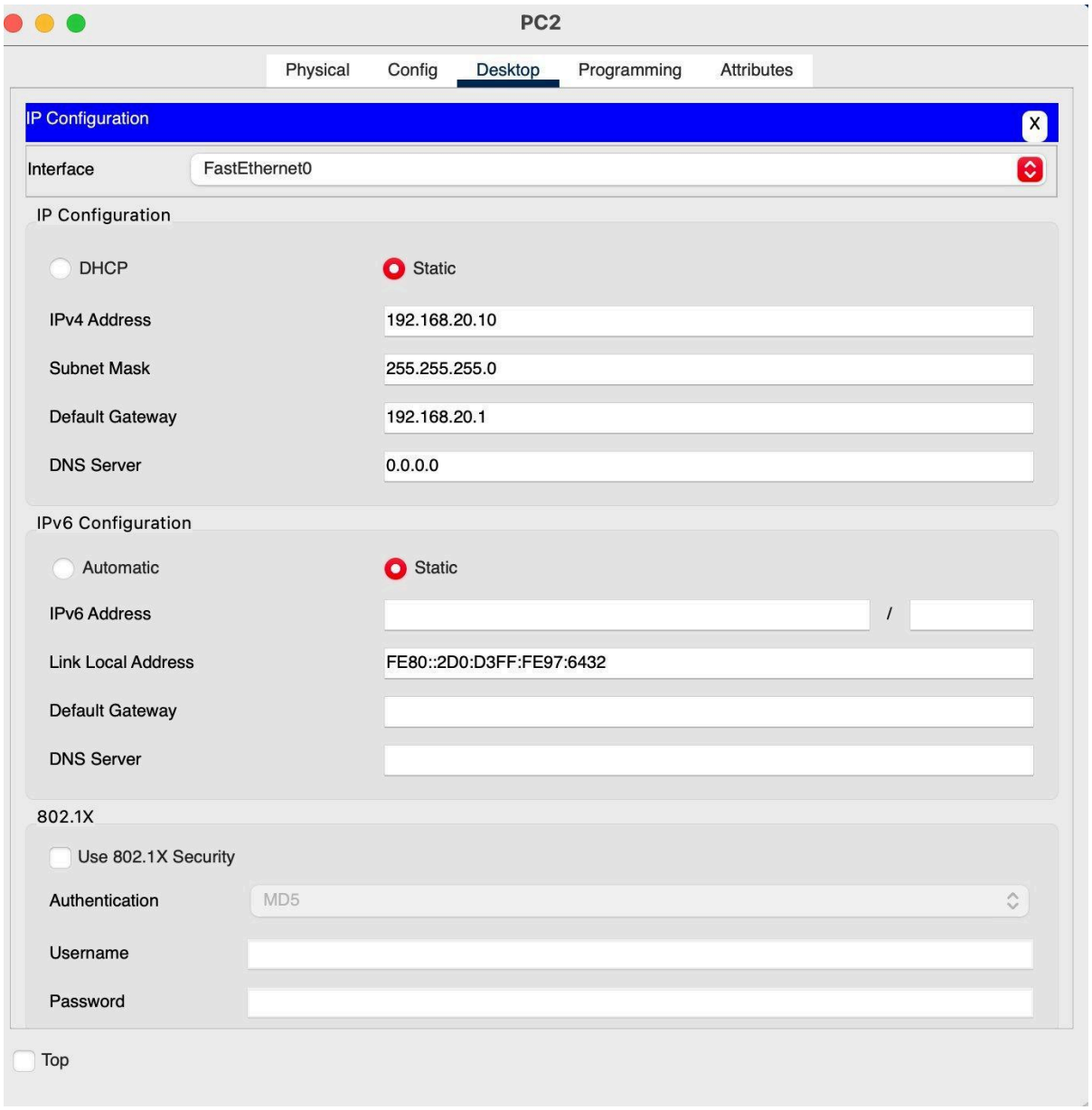
Each PC was assigned a static IP address, subnet mask, and default gateway corresponding to its VLAN.

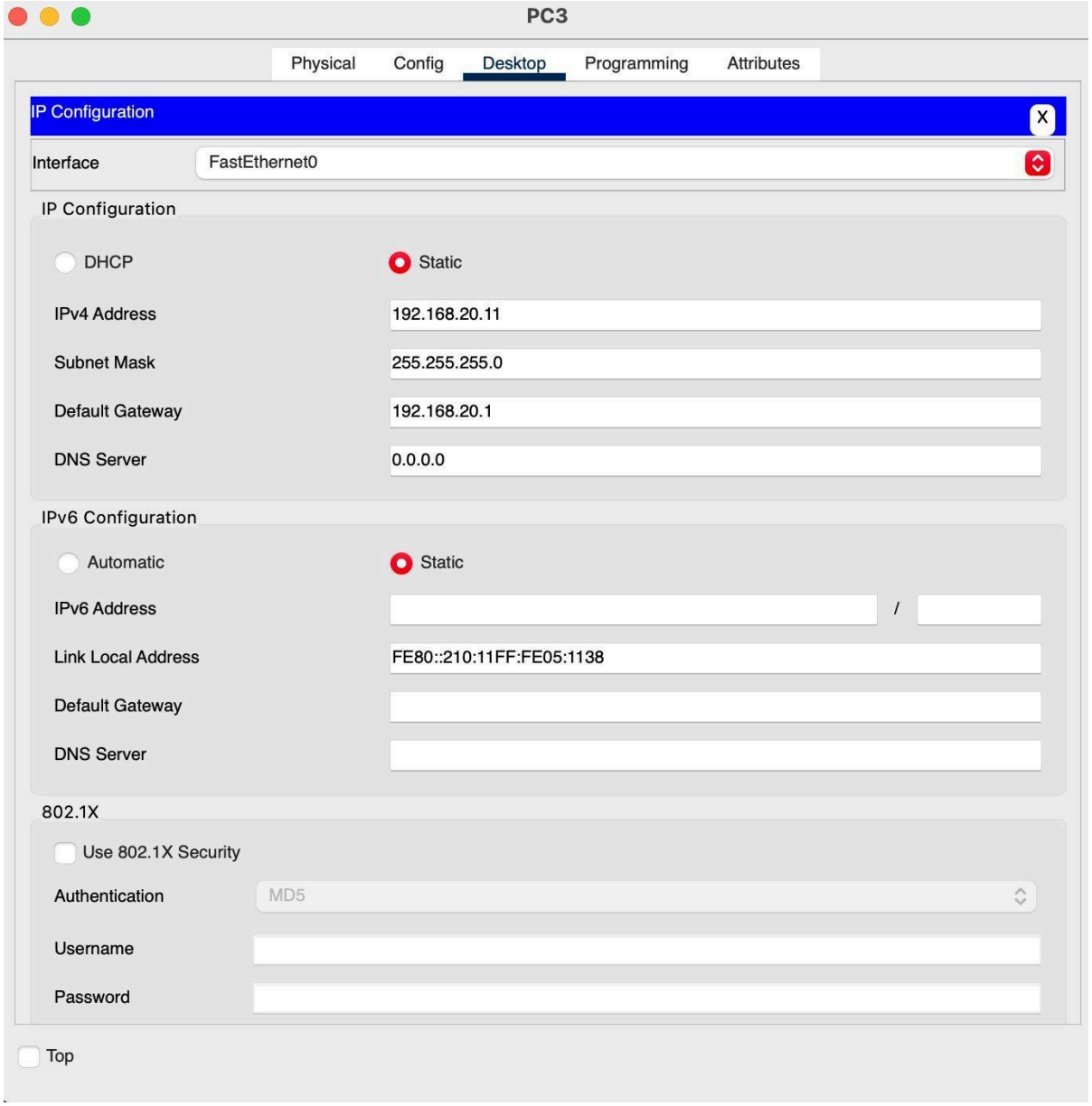
VLAN 10 — Students


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::200:CFF:FEE8:8974
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2D0:FFFF:FEC6:6222
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

VLAN 20 — Faculty

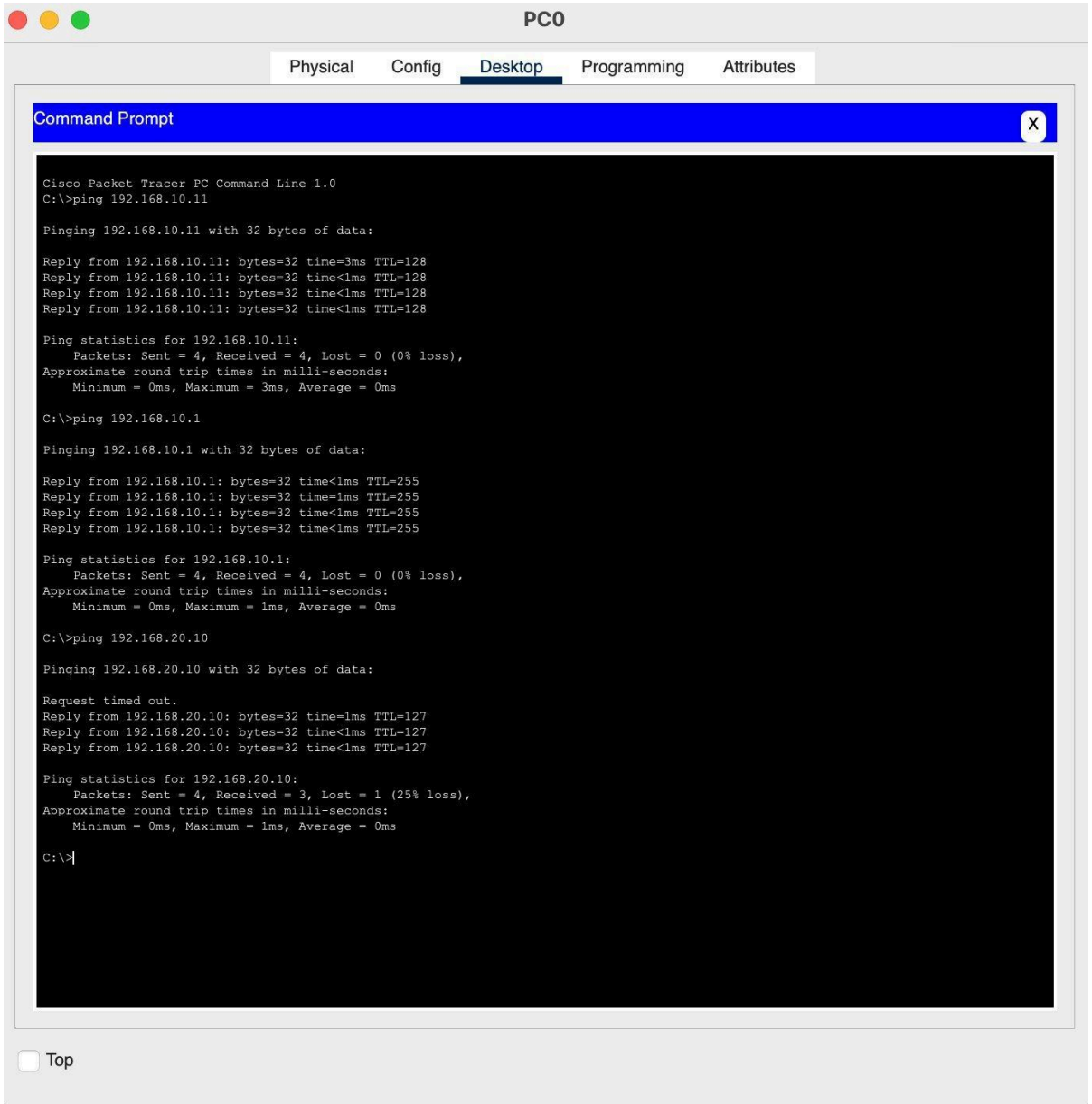




7. Connectivity Testing

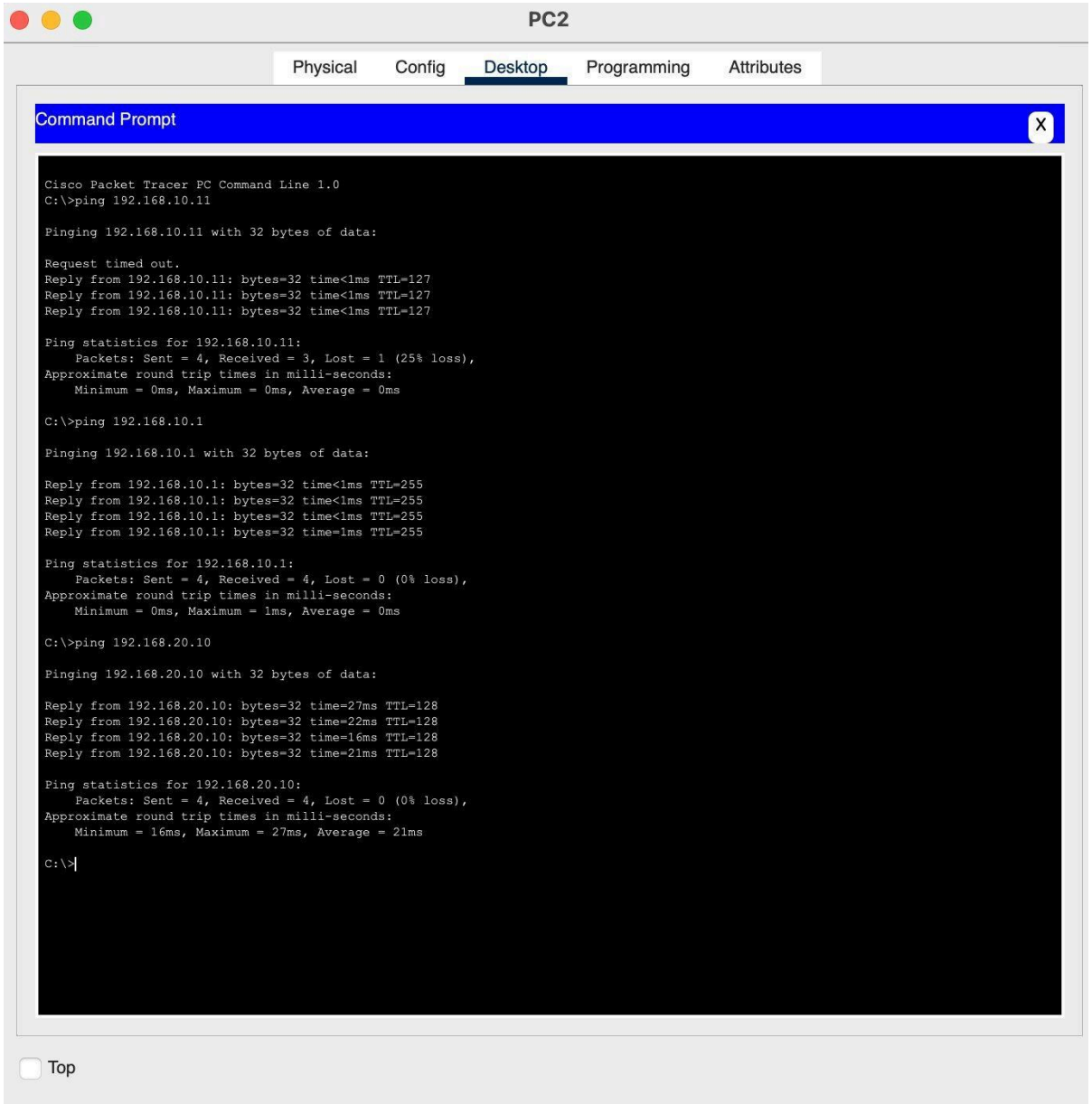
Intra-VLAN and Inter-VLAN Ping Tests (PC0)

Pings were sent from PC0 (VLAN 10) to PC1 (same VLAN) and PC2 (VLAN 20). Intra-VLAN ping was fully successful. The initial inter-VLAN ping had one timeout due to ARP resolution, with subsequent replies succeeding. TTL of 127 confirms routing through the router.



Inter-VLAN Ping Tests (PC2)

From PC2 (VLAN 20), pings were sent to both the gateway and a host in VLAN 10 (192.168.10.11), confirming bidirectional inter-VLAN communication.



8. Observations

The following key observations were made during the lab:

1. Initial inter-VLAN pings failed because the switch-to-router link was not configured as a trunk. When PC0 (VLAN 10) attempted to reach a host in VLAN 20, the traffic could not traverse Fa0/1 as VLAN-tagged frames were not being carried.
2. Trunking (802.1Q) resolved the inter-VLAN communication issue. Configuring switchport mode trunk on Fa0/1 allowed both VLAN 10 and VLAN 20 traffic to reach the router over the same physical link.

3. Router subinterfaces enabled routing between VLANs once trunking was active. Subinterfaces G0/0.10 and G0/0.20 with encapsulation dot1Q correctly received tagged traffic and routed between networks.
4. ARP caused occasional first-ping timeouts after configuration was correct. Subsequent pings succeeded consistently once MAC addresses were learned.
5. The routing table verified correct Layer 3 configuration. show ip route confirmed both networks were directly connected via the router subinterfaces.

9. Conclusion

This lab successfully demonstrated VLAN implementation and network segmentation. By configuring VLANs on the switch, assigning ports, enabling 802.1Q trunking, and configuring router-on-a-stick subinterfaces, full inter-VLAN routing was achieved.

Connectivity tests confirmed proper segmentation within VLANs, successful communication between VLANs via the router, correct routing table entries, and proper trunk encapsulation. The objectives of Lab 2 were fully met.